Om Patel

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Education

North Carolina State University — Raleigh, NC

Master of Science in Computer Engineering — *May 2028 | GPA: 4.0*

Bachelor of Science in Computer Engineering — *May 2027 | GPA: 3.98*

- Coursework: Parallel Computer Architecture, Digital ASIC and FPGA Design, Microprocessor Architecture, Gen AI for Computer Systems, Embedded Systems

Technical Skills

Tools: AMD Vivado, Icarus Verilog, Git/GitHub, JSON, KiCAD, GTKWave, PyTorch, Linux, Docker, VS Code
Technologies : FPGA, SPI, OpenMP, RISC-V, I2C, UART, CAN, HPC, MAX78000, OpenAI/Ollama APIs
Languages: C, C++, Python, Java, HTML/CSS, Verilog, SystemVerilog, VHDL, MatLab, SQLite, Assembly

Experience

FPGA Engineer Intern — May 2026 – Present

Lockheed Martin (previously Ultra Maritime) — *Wake Forest, NC*

- **Implemented** a bidirectional **SPI controller in VHDL** (master and slave modes), enabling full-duplex communication between an FPGA and microcontroller at 460.8 kbps, supporting GUI-driven configuration and data exchange.
- **Utilized asynchronous FIFOs** to handle **clock domain crossing** between the FPGA system clock and ADC sampling clock domains in **SystemVerilog**, preventing metastability and ensuring reliable data transfer across asynchronous boundaries
- **Optimized** statistical post-processing (mean, standard deviation) and scaling operations on FFT output for real-time signal analysis on FPGA hardware
- **Wrote testbenches** and performed **functional simulation** across SPI, CDC, and DSP modules to verify correctness prior to hardware integration, catching timing and logic issues early

Technical Projects

Systolic Array Matrix Multiplier | *Verilog, Python, Icarus Verilog*

- **Designed and implemented** a **4×4 output-stationary systolic array** in Verilog for matrix multiplication, verified bit-exact against a numpy golden referenc
- **Built** a tiled inference pipeline that classifies 8×8 handwritten digit images using **int8-quantized weights**, achieving **95.83% accuracy** with zero degradation from quantization
- **Optimized** tile scheduling with double-buffered loading, overlapping data transfer and computation to **reduce inference latency from 1,548 to 890 cycles** (42.5% improvement)
- **Wrote** unit and integration testbenches verifying correctness across 10 test images with bit-exact score matching against the Python reference model

RISC-V Pipelined Processor | *Verilog, RISC-V, Vivado*

- **Designed** a **5-stage pipelined 32-bit RISC-V processor** in Verilog (IF → ID → EX → MEM → WB) supporting the RV32I base integer instruction set
- **Implemented** data forwarding logic and pipeline stall/flush mechanisms to resolve data hazards (RAW) and control hazards (branch misprediction)
- **Achieved 100 MHz clock frequsency** post-synthesis on a Basys3 Artix-7 FPGA, a **54% improvement** over the single-cycle baseline of 65 MHz
- **Verified** pipeline correctness across **15 test programs** covering ALU operations, load-use hazards, back-to-back dependencies, and branch sequences

LLM-Guided Cache Optimization | *Python, C++, SQLite3, JSON*

- **Built** an **LLM-driven framework** in **Python** combining FunSearch and Graph of Thought reasoning to evolve CPU cache replacement policies, storing results in JSON and **analyzing performance using SQLite3**
- **Achieved** a **22% improvement in mean IPC** compared to baseline SRRIP and SHiP algorithms through optimized policy evolution and ablation analysis, placing 5th among 50 undergrad and grad students